

Agile & Scrum for QA Engineers

What we learn today

- Why Agile?
- Agile Manifesto & Principles
- Agile vs Waterfall
- Key Agile methodologies (Kanban, Scrum)
- Scrum Framework with AI tooling
- Scrum Roles in AI-enabled teams
- Scrum Ceremonies with AI assistants
- Sprint Life-cycle
- AI Augmented Development Workflow

Why Agile?

Previously, most software was developed using the **Waterfall Model**. However, it had a major problem.

For example, suppose a software project takes **6 months** to develop. At the end, the client says:

I don't need this feature anymore. I need a different feature.

At that point, changing the entire system becomes very **costly and difficult**.

To solve these problems, **Agile** was introduced.

Main Objectives of Agile

- Deliver software faster
- Easily adapt to changing requirements
- Get regular customer feedback
- Complete work in small, manageable increments
- Find and fix bugs quickly

Benefits of Agile

- Faster Delivery
- Better Customer Satisfaction
- Easy to Handle Changes
- Continuous Testing
- Reduced Risk

Agile Manifesto

There are 4 core values:

- Individuals and Interactions
- Working Software
- Customer Collaboration
- Responding to Change

There are 12 agile principles:

- Customer Satisfaction First
- Welcome Changing Requirements
- Deliver Frequently
- Developers & Business Work Together
- Build Around Motivated People
- Face-to-Face Communication
- Working Software is Progress
- Sustainable Development
- Technical Excellence
- Simplicity
- Self-organizing Teams
- Continuous Improvement

Agile vs Waterfall

Aspect	Waterfall	Agile
Process	Sequential	Iterative
Requirements	Fixed at the beginning	Can change throughout the project
Delivery	Delivered once at the end	Delivered in small, frequent releases
Testing	Testing is mainly done at the end	Testing is performed in every Sprint
Customer Feedback	Usually received at the end	Received continuously
Risk	Higher	Lower
Flexibility	Low	High
QA Involvement	QA is mainly involved after development	QA is involved from the beginning

Key Agile Methodologies (Kanban, Scrum)

Agile is not a single methodology. It is an approach that includes several frameworks and methodologies.

The most popular ones are:

Scrum

One of the most widely used Agile frameworks.

Key Features

- Sprint
- Product Backlog
- Sprint Backlog
- Daily Stand-up
- Sprint Review
- Sprint Retrospective

Kanban

Kanban is a visual workflow management method.

Board Example

To Do



In Progress



Testing



Production

Kanban does not require Sprints.
It follows a continuous flow of work.

Sprint Vs Retrospective

Sprint Review → Focus on the Product

We have completed the Money Transfer Feature. Users can now transfer money from one account to another.

The team discusses:

- Is the feature working as expected?
- Are the requirements fulfilled?
- What is the stakeholder/customer feedback?
- Are any changes needed?

Example: Demo the Money Transfer Feature.

Sprint Retrospective → Focus on the Team & Process

During the Sprint, the Developer completed the feature and communicated the changes properly with the QA team. As a result, QA was able to start testing on time.

However, the requirements were provided late, so QA had less time to complete testing.

The team discusses:

- What went well?
- What went wrong?
- What can we improve in the next Sprint?

Example: Discuss how to build requirements in the next Sprint.

Scrum vs Kanban

Scrum	Kanban
Uses Sprints	No fixed Sprints
Work is organized into fixed time periods	Continuous Flow
Defined Roles	Flexible Roles
Sprint Planning is Required	Continuous Planning

Kanban



Sprint



AI-Augmented Development

AI-Augmented Development = Human expertise + AI assistance = Faster & smarter software development.